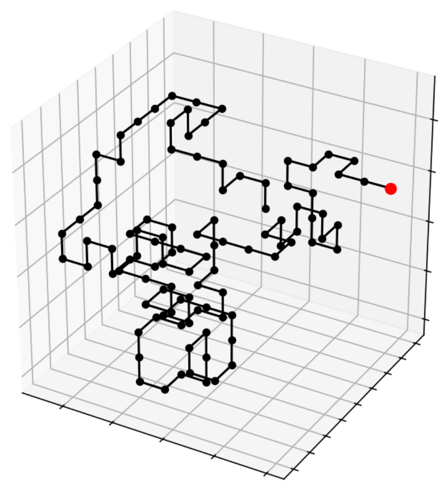

Cartesian and Spherical 3-dimensional Random Walks

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January 26, 2026

Course: Simulation and Modeling - SI1336
Royal Institute of Technology

Contents

1 Introduction

1.1 Background

In this case, we define a random walk in $\mathbb{R}^n$ of length N as

$$\boldsymbol{u}_{i+1} = \boldsymbol{u}_i + \boldsymbol{\Delta u}_i, \quad \boldsymbol{u}_0 = \boldsymbol{v}, \quad i = 0, 1, ..., N-1,$$

where $\boldsymbol{\Delta u}_i$ is the direction and the distance we take in each step of the walk and $\boldsymbol{v}$ is the starting point of the walk. The walk set $U_N = \{\boldsymbol{u}_0, \boldsymbol{u}_1, ..., \boldsymbol{u}_N\}$ contains all the nodes of the walk, with each node being connected to its adjacent neighbor.

1.2 Objective

In this project, the objective is to analyze and compare qualitative and quantitative behaviors of different types of walks in $\mathbb{R}^3$. It is also in our interest to see how the behaviors of the walks are affected by putting certain limitations and rules to their movement pattern.

In particular, we will be comparing random walks with cartesian and spherical steps by comparing the distribution of their reaching distances. We will also compare the different types of walks with themselves when they have different constraints on the regions for which the next step will be randomized.

2 Methods

2.1 Cartesian Walk

For the setup of this study, we define a cartesian random walk in $\mathbb{R}^3$ of length N as

$$\boldsymbol{u}_{i+1} = \boldsymbol{u}_i + \boldsymbol{e}_i, \quad \boldsymbol{u}_0 = \boldsymbol{0},$$

where $\boldsymbol{e}_i \in \{\pm\boldsymbol{e}_1, \pm\boldsymbol{e}_2, \pm\boldsymbol{e}_3\}$ is a cartesian step in an axis-parallel direction. All the outcomes of $\boldsymbol{e}_i$ in each step of the walk are as likely. A modification of this walk is to limit its movement by not letting it collide with itself in the last step, meaning that

$$\boldsymbol{u}_{i+1} \neq \boldsymbol{u}_{i-1}.$$

One can also modify the walk further so that each node in U_N is unique, meaning that the walk never collides with itself. This can be expressed as

$$\boldsymbol{u}_{i+1} \notin U_i = \{\boldsymbol{u}_0, \boldsymbol{u}_1, ..., \boldsymbol{u}_i\}.$$

2.2 Spherical Walk

For the setup of this study, we define a spherical random walk in $\mathbb{R}^3$ of length N as

$$\boldsymbol{u}_{i+1} = \boldsymbol{u}_i \pm \boldsymbol{r}_i, \quad \boldsymbol{x}_0 = \boldsymbol{0},$$

where $\boldsymbol{r}_i = \boldsymbol{r}_i(\theta, \phi)$, $\theta \in [0, \pi]$ and $\phi \in [0, 2\pi]$ is a spherical unit vector with the polar angle θ and the azimuthal angle ϕ. All the outcomes of $\boldsymbol{r}_i$ in each step of the walk are as likely. A modification of this walk is to limit its movement by setting a lower limit to the angle the step vectors can for with its predecessor. Let this angle be θ and define the constraint

$$\theta_{min} \leq \theta = \pi - arccos(\boldsymbol{r}_{i+1} \cdot \boldsymbol{r}_i),$$

where θ_{min} is the minimum angle that we allow between the steps of the spherical walk.

2.3 Statistical Approaches

The resulting data will be based on a large number of simulations. To analyze it we will have use of the arithmetic mean $\bar{x}$. For a set $\{x_i\}_{i=0}^N$ this is given by

$$\bar{x} = \frac{1}{N} \sum_{i=0}^{N} x_i. \tag{1}$$

Because the experiments are based on samples, we will calculate the standard deviation by using

$$s = \sqrt{\frac{1}{N-1} \sum_{i=0}^{N} (x_i - \bar{x})^2}. \tag{2}$$

From equation (2) we can finally derive the standard error of the sample as

$$SE = \frac{s}{\sqrt{N}} = \sqrt{\frac{1}{N(N-1)} \sum_{i=0}^{N} (x_i - \bar{x})^2}. \tag{3}$$

3 Experiments and Results

3.1 Cartesian Walk

For the cartesian walk we first analyze the reaching distance for walks of different lengths. It is also of interest to compare this walk to its variations with restrictions, which in this case is a walk that avoids its previous node, and one that avoids all previous nodes. With Equations (1) and (3), this comparison is given i Figure (1).

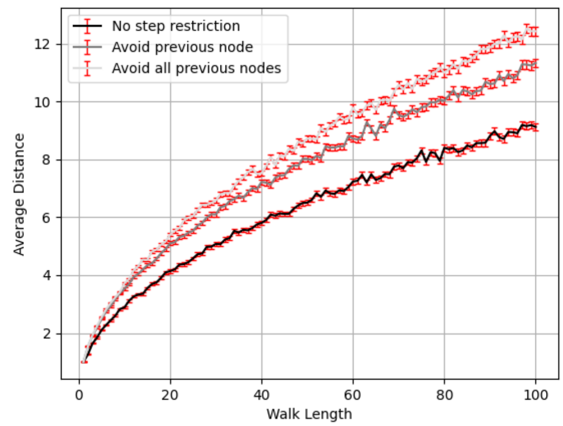

Figure 1: Reaching distance for cartesian walks of different lengths up to 100 and different restrictions on the steps, averaged over 1000 simulations.

We now study the rate of walks that have completely unique nodes for each step method, meaning that we look at the fraction of walks that are self-avoidant over the total amount of walk. This result is given in Figure (2).

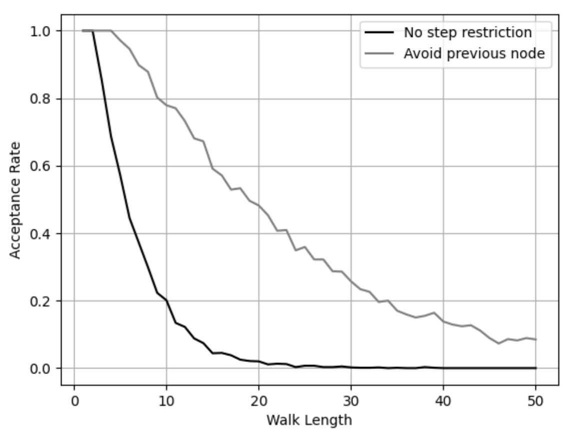

Figure 2: Acceptance rates for the non restricted cartesian walk and the cartesian walk avoiding its predeceasing node, for different lengths up to 50 run with 1000 samples per length.

The used method for obtaining a walk that is guaranteed to have unique nodes is based on a recursive depth-first search for slots in the space that are not occupied bu other nodes. the depth over each iteration for this algorithm for a few walks of length is demonstrated in Figure (3). One might then be interested in analyzing the iteration to length ratio, to determine how effective it becomes for large lengths. With Equations (1) and (3), this is analysis is given in Figure (4).

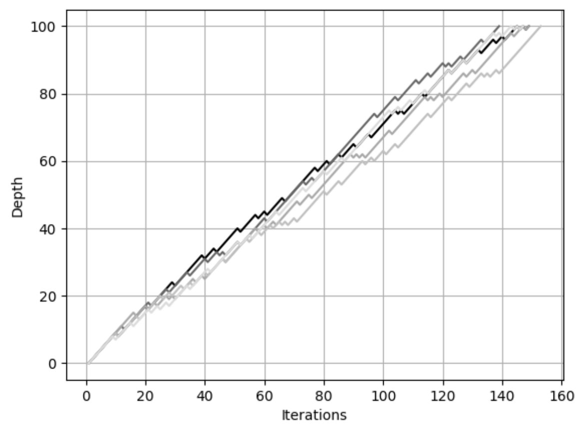

Figure 3: Examples of depth progression for the cartesian walk avoiding all previous nodes over its iterations.

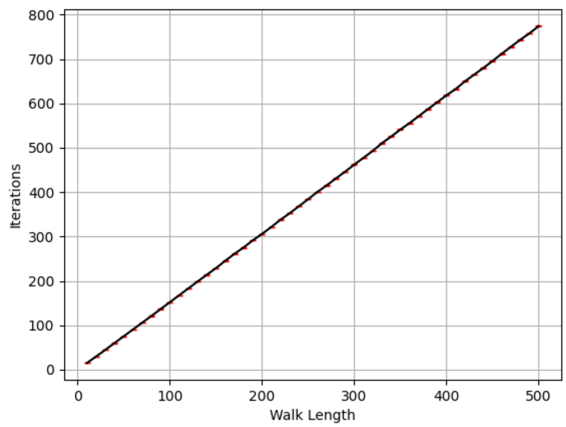

Figure 4: The number of iterations for the cartesian walk avoiding all previous nodes depending on length, for lengths up to 500 run with 1000 samples per length.

3.2 Spherical Step

For the spherical walk we first analyze the reaching distance for walks of different lengths. It is also of interest to compare this walk to its variations with restrictions, which in this case is a walk that has a minimum angle that we allow between the steps of the spherical walk. With Equations (1) and (3), this comparison is given i Figure (5).

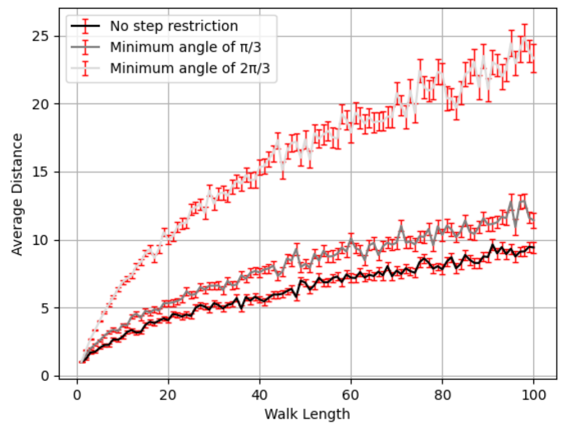

Figure 5: Reaching distance for spherical walks of different lengths up to 100 and different restrictions on the steps, averaged over 100 simulations.

If we keep the length constant and present this in a different way using gaussian kernel density estimates, we can represent the distribution of lengths, for more different angle restrictions in Figure (6).

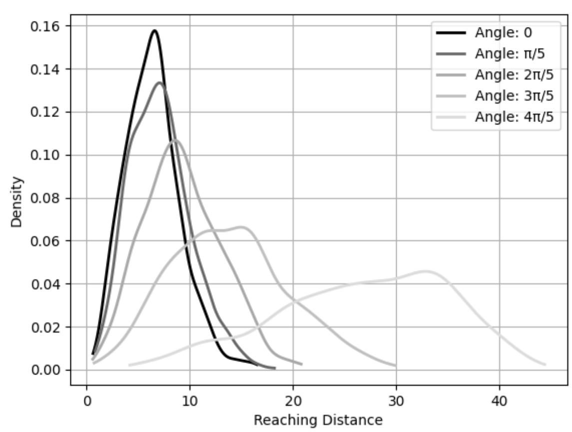

Figure 6: Density distribution of reaching distance for different minimum angles in each step of the spherical walk, with a fixed length of 50 and 1000 samples per angle setting.

3.3 Comparison

We are lastly interested to look at the difference of the cartesian and spherical walk, both unrestricted in their movement. Using equations (1) and (3), both plots together are given in Figure (7).

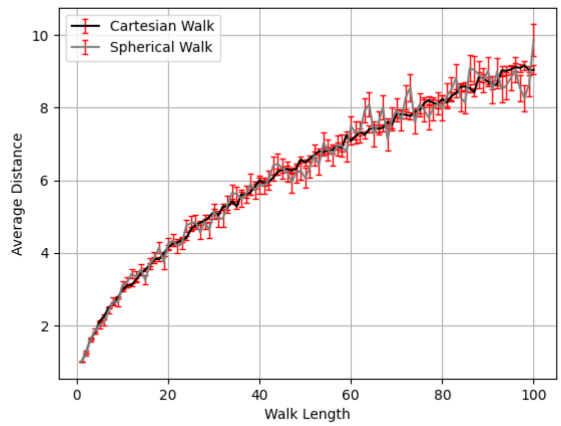

Figure 7: Reaching distance for cartesian and spherical walks of different lengths up to 100, averaged over 1000 simulations.

4 Discussion

We can clearly see in Figure (1) that the reaching distance om each cartesian walk has a significant difference, where the one that never collides with itself will have the largest reaching distance. However, since the reaching distance of a random walk is given by $\sqrt{u_{Nx}^2 + u_{Ny}^2 + u_{Nz}^2}$ (where u_{Nx}, u_{Ny} and u_{Nz} are the cartesian components of u_N), we get that the distance is proportional to $f(x) = \sqrt{k + x^2}$, $k \in \mathbb{R}$, in each direction. For large distances the walks will instead be approximately be proportional to $f_{far}(x) = x$ in each direction, meaning that the only difference between them will be a constant term. The walks should therefore theoretically inherit the same behavior at large distances, but to further confirm this one should conduct numerical experiments of this for large lengths, which has not been possible here due to limitations of computational power.

Regarding the acceptance rates of successful cartesian walks completely avoiding themselves, we can see in Figure (2) that the number of walks that are accepted in the case of the walk avoiding the last node is greater than in the non-restricted walk. This is to be expected since 1/6 of the non-restricted walks get discarded because of a collision with the previous node, which is not an option with the other walk.

Observe that there is no point in studying the walk-method that avoids all previous nodes, since it will always have an acceptance rate of 100%. However, it is of great interest to see how it progresses in its depth over iterations, and as we can see in Figure (3), the behavior seems to be rather constant. More rigorously, the iterations I necessary to obtain a walk of length L can, with good accuracy as seen in Figure (4), be linearly modeled as $I(L) = 3L/2$.

Ultimately, if one is looking to achieve a short random walk with unique nodes, the method of avoiding the previous node (and discarding walks that do not have unique nodes) and the method of avoiding all nodes will be of good choice, while for longer walks the latter will be the best option.

When analyzing the spherical walk, one must be careful with the method of choice. If we choose to randomize a polar angle θ and an azimuthal angle ϕ to then use polar coordinates to obtain our step, we will get a bias at $\theta = 0, \pi$ which makes the steps nonuniform. This phenomenon is shown in Figure (8). One easy fix to this however is to, instead of randomizing angles, randomize points in the box $\{(x, y, z); x, y, z \in [-1, 1]\}$ and then normalize points within the unit sphere and discard points outside of it (this in order to avoid a bias towards the corners of the box). Doing this, we get a uniform distribution of our steps, which is demonstrated in Figure (9).

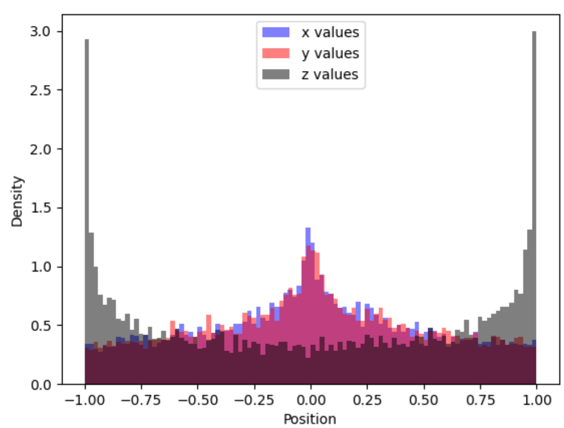

Figure 8: Histogram of cartesian component values for 10000 randomized spherical steps, using the method of randomizing a polar angle θ and an azimuthal angle ϕ.

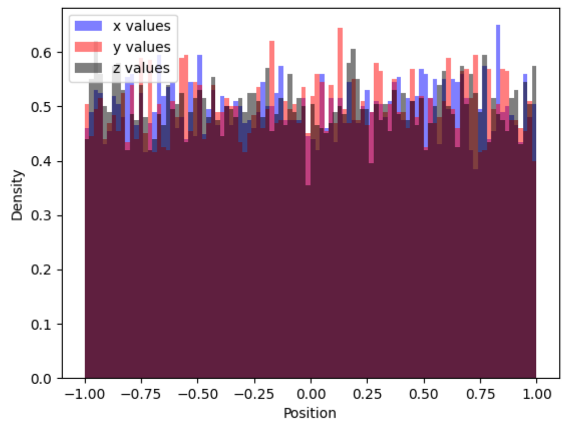

Figure 9: Histogram of cartesian component values for 10000 randomized spherical steps, using the method of randomizing a point in the box $\{(x, y, z); x, y, z \in [-1, 1]\}$.

When we analyze Figure (5) we get the expected behavior that a larger angle restriction leads to longer reaching distances, since the spherical walk cannot make as sharp turns in each step. Figure (6) demonstrates this in the way that the distribution of the reaching distances converges to the length of the walk, which intuitively makes sense. Unfortunately, due to limitations in computational power, the spherical walks could not be evaluated with the same accuracy as for the cartesian walk.

When comparing the unrestricted cartesian and spherical walk, we can see that they have similar behavior. However, the error of the spherical walk is greater, most likely because it has more degrees of freedom to move in, while the certainty is better for the cartesian walk. In the unrestricted case this means that the cartesian walk can be a good supplement for the spherical walk, trading of degrees of freedom for less uncertainty.